

Pig Detection Method Based on Improved YOLOv5

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Abstract—The inevitable path to addressing the expanding scale of pig farming and the recurring swine fever outbreaks in China lies in the intelligentization of pig farms, with the accurate identification of individual pigs serving as a cornerstone. Despite recent advancements where numerous researchers have integrated deep learning with animal husbandry, proposing various object detection techniques, these methods have fallen short in practical applications within pig farms. We identified that these techniques often suffer from slow processing speeds or inadequate detection accuracy, particularly due to pig occlusion. To tackle these challenges, we introduce an enhanced pig detection approach based on an improved YOLOv5. By incorporating BiFPN and Ghost modules into the conventional YOLOv5 algorithm, we've managed to boost detection accuracy without compromising the model's size. Additionally, we've incorporated a counting function that instantly displays the number of pigs detected on images or videos, facilitating effortless enumeration. Experimental results have confirmed the efficacy of our proposed algorithm.

Keywords—pigs; YOLOv5; counting; object detection; Deep learning)

I. INTRODUCTION

Pork is the most widely consumed meat in China, known for its mild and slightly sweet flavor along with its high nutritional value. The pig farming sector is a key and rapidly growing part of the country's livestock industry [1]. China holds the position as the largest producer and consumer of pork globally. According to FAO data, global pork production is projected to reach 124.5 million tons in 2023, with China contributing 58.8 million tons, which represents 47% of total worldwide output. The pig industry encompasses various stages, including research and development, feed production, breeding, commercial pig farming, meat processing, cold chain logistics, and food service retail, impacting the sustainability and growth of numerous related industries. This underscores the significant role pigs play in the economy.

As pig farming in China continues to expand and outbreaks of swine fever become more common, the adoption of intelligent management systems in breeding farms has become essential [2]. A critical aspect of this transition is the accurate identification of individual pigs, which plays a vital role in disease prevention, pork quality traceability, and shaping the animal husbandry insurance sector. Traditionally, managers have relied on methods such as color tags, tattoos, and RFID tags for pig identification [3].

However, RFID systems have limitations, including a restricted operational range and susceptibility to interference from other electromagnetic devices, which can compromise communication between tags and readers [4]. This not only results in inefficient use of labor but can also induce stress in pigs, raising concerns about animal welfare and leading to financial losses for farmers.

To tackle these challenges, more researchers are exploring the integration of computer vision and deep learning technologies into pig farming. Despite the development of numerous object detection techniques, problems like target movement and occlusion continue to hinder recognition accuracy, preventing these methods from being effectively implemented on actual pig farms. Consequently, developing a reliable and efficient approach for pig detection is crucial and holds significant practical value.

Object detection is a crucial area within computer vision that involves not only identifying the location and size of objects but also categorizing them within images. This field is typically classified into One-Stage detection algorithms, like YOLO and SSD, and Two-Stage detection algorithms, such as Faster R-CNN. One-Stage algorithms predict both the category and position of objects directly from the image, whereas Two-Stage algorithms first propose candidate regions before classifying and refining them with bounding boxes. For the purpose of achieving real-time and efficient pig detection, this study adopts the faster One-Stage detection approach and enhances it using the YOLOv5 model. The key contributions of this research are outlined as follows:

1. Change the network structure of YOLOv5. In the Head part, some small filters are used to generate more feature maps from the original convolutional layer to achieve better recognition performance.

2. The network structure of YOLOv5 is changed. In the Backbone part, bidirectional connections are used to better capture feature representations at different scales, improve the ability of the network to deal with objects of different sizes and complexity, and then improve the robustness and accuracy of detection.

3. Add the counting function to show the number of detected pigs in the detection results for convenient statistics.

4. It is verified on the self-built dataset and achieves better recognition performance.

This article is organized as follows:

Sections I and II offer a brief overview of the relevant research and historical development of object detection. Section III presents an in-depth explanation of the methodology. Section IV covers the dataset utilized, along with the experimental results and comparative analysis. Finally, Section V concludes with a summary of the method.

II. RELATED WORK

Object detection is a key area within computer vision focused on identifying and locating particular objects in images or videos. The advancement of deep learning technologies has greatly enhanced the accuracy and speed of object detection algorithms, establishing them as a crucial foundation for numerous real-world applications.

Traditional object detection methods primarily focus on manually extracting features like SIFT and SURF. These approaches rely heavily on specific feature descriptors for object recognition but struggle with complex backgrounds and variations in lighting. The emergence of deep learning technology has led to significant advancements in object detection algorithms. Before 2012, algorithms were primarily traditional; however, following the introduction of AlexNet in 2013, a shift to deep learning techniques occurred. In 2014, R. Girshick and colleagues introduced the RCNN algorithm [5], which applied convolutional neural networks (CNN) to object detection, significantly enhancing performance and reshaping research directions in the field. The following year, He et al. developed the SPPNet algorithm [6], which extracts regional features from feature maps after convolution and does not require a fixed input size. In 2016, Redmon et al. introduced the YOLO algorithm [7], marking a pivotal moment in object detection by addressing the speed and efficiency issues of traditional methods, and classifying detection into One Stage and Two Stage approaches. That same year, Liu et al. presented the SSD algorithm [8], which performs regression on feature maps of varying depths to enhance detection accuracy for both large and small objects. In 2017, Lin et al. proposed the RetinaNet algorithm [9], which utilizes Focal Loss to tackle class imbalance and further improve object detection accuracy.

To promote intelligent aquaculture and address identity recognition challenges, an increasing number of researchers are integrating object detection technologies. Wang et al. enhanced the YOLOv3 algorithm by incorporating residual modules into the Darknet-53 network, which boosted the recognition capability of cow targets in agricultural settings [10]. Additionally, Wang et al. employed the widely-used Faster R-CNN network structure from deep learning classification algorithms, utilizing a yak dataset and sample data from ImageNet. This approach transformed scene object detection into a binary classification problem, thereby improving yak detection efficiency [11]. Chen implemented the CBAM attention module within the feature extraction network to enhance the acquisition of comprehensive image information, resulting in improved accuracy and real-time performance [12]. Liu advanced the YOLOv5 network by focusing on multi-scale feature fusion, leading to the development of a new object detection algorithm called

DFD-YOLOv5, which demonstrated superior recognition capabilities [13].

III. METHODOLOGY

A. YOLOv5

The YOLO algorithm has significantly advanced the speed and accuracy of object detection. It streamlines the detection process by transforming the object detection challenge into a regression problem, thereby eliminating the need for the complex region extraction and post-processing typically associated with traditional methods. This approach simplifies implementation and optimization, lowering development complexity. Its effectiveness and precision have led to its widespread application in numerous real-world scenarios, contributing to the progress of computer vision and establishing a foundation for many subsequent object detection models, while also fostering ongoing enhancements in real-time object detection technology.

The YOLOv5 algorithm incorporates a lighter network architecture and utilizes more efficient convolution modules, such as CSPDarknet53 [14] and SPP [15], to enhance feature extraction efficiency. It also integrates automated data augmentation methods like Mosaic data augmentation [16] and MixUp [17], which increase the model's robustness against varied data inputs. Additionally, novel loss function designs, such as CloU [18], have been implemented to boost the precision of bounding box regression. More effective training strategies and optimizers, including the Adam optimizer [19] and a learning rate scheduler, have been introduced to enhance training stability and convergence speed. Overall, YOLOv5 has achieved notable advancements in detection accuracy, training efficiency, and inference speed by combining multiple cutting-edge technologies, making it more competitive for practical applications.

B. Our overall network structure

In recent years, numerous object detection methods have emerged; however, challenges like object movement and occlusion still result in reduced recognition accuracy, limiting their practical application in pig farms. Additionally, many of these methods exhibit low efficiency, often leading to target loss due to the slow post-processing speed that follows the capture of real-time image data.

To address these issues, we have enhanced the YOLOv5 algorithm by substituting the C3 module with a more efficient Ghost [20] module, which lowers computational complexity while maintaining accuracy, thus achieving a more lightweight design. Additionally, we replaced the Concat module with the BiFPN [21] module, whose bidirectional connections improve the capture of feature representations at various scales, enhancing the model's efficiency and performance. By implementing these technical methods, we enable real-time presentation of results on images or videos, thereby boosting the efficiency of farm operations and the algorithm's real-time capabilities. The subsequent sections of this chapter will provide a comprehensive overview of these enhancements. The overall structure of our network is illustrated in Figure

1

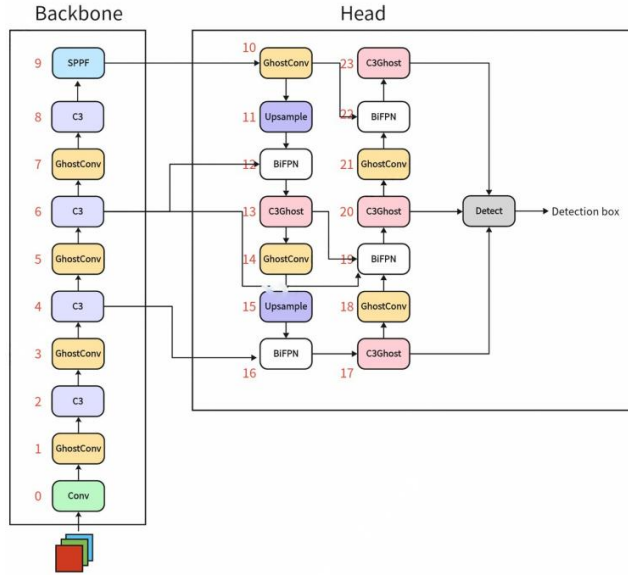


Figure 1. Overall network structure.

C. Module Ghost

Implementing convolutional neural networks on embedded devices poses challenges due to constraints in memory and computing power. The effectiveness of feature maps is a key characteristic of successful convolutional neural networks, as illustrated by a specific network's feature map in Figure 2. The feature maps represented by the red, green, and blue boxes in the figure are quite similar and can be derived through simple linear transformations rather than complex non-linear ones. This suggests that one feature map can serve as a "Ghost" of another. Consequently, not all feature maps need to be generated through convolution; instead, "Ghost" feature maps can be created using less resource-intensive operations. This is why we have incorporated the Ghost module in YOLOv5 to maintain accuracy while achieving a lightweight design.

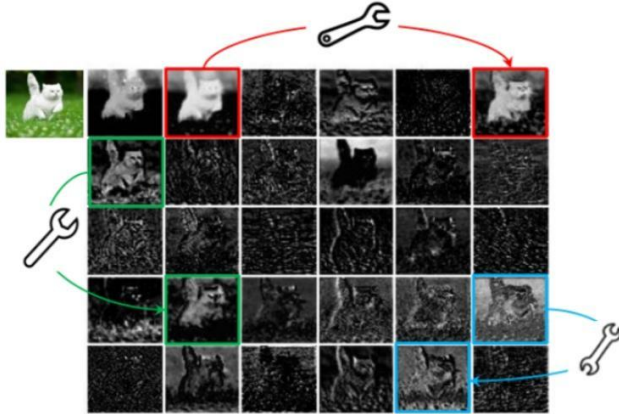


Figure 2. A network feature map [20].

The Ghost module, illustrated in Figure 3, begins by executing a limited convolution to decrease computational

complexity. The next step involves performing a group convolution operation (where ϕ_i represents a linear transformation) to derive the red section of the feature map beneath the output feature map. Finally, the use of Identity allows for the summation of the channel counts obtained from the first two steps. When compared to deep convolutional neural networks, the Ghost module offers a notable advantage: if the output of n feature maps is known, it has an identity mapping (i.e., m feature maps are produced through standard convolution) and performs $m(s-1) = (s-1)n/s$ linear operations (where $n=ms$). Each linear operation employs a convolution kernel size of $d \cdot d \cdot d$. Thus, it can be concluded that the Ghost module and standard convolution can theoretically generate the same number of feature maps while maintaining equivalent acceleration and parameter compression ratios. Assuming $k=d$ and s is much smaller than c , the theoretical acceleration ratio is:

$$r_s = \frac{n \cdot h' \cdot w' \cdot c \cdot k \cdot k}{\frac{n}{s} \cdot h' \cdot w' \cdot c \cdot k \cdot k + (s-1) \cdot \frac{n}{s} \cdot h' \cdot w' \cdot d \cdot d} \quad (1)$$

$$r_s \approx \frac{s \cdot c}{s + c - 1} \approx s \quad (2)$$

Consequently, the Ghost Module can theoretically reduce training time by a factor of s and decrease the number of parameters by the same factor. (The hyperparameters s and K are adjustable, with s controlling the number of feature maps in conventional convolution methods, while dd refers to the size of the convolution kernel used for linear transformations.)

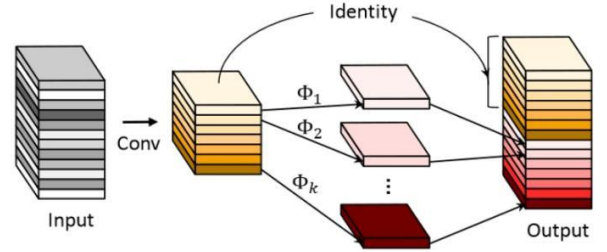


Figure 3. Module Ghost [20].

D. Module BiFPN

To enhance the accuracy of the algorithm, we incorporated the BiFPN module into YOLOv5. BiFPN, which stands for Bidirectional Feature Pyramid Network, facilitates rapid and efficient multi-scale feature fusion. The structure of BiFPN is illustrated in Figure 4.

The working principle of BiFPN is summarized as follows:

1. Feature Pyramid Generation: Initially, the network generates a feature pyramid by extracting features from multiple layers of the backbone network (usually convolutional neural networks such as ResNet).

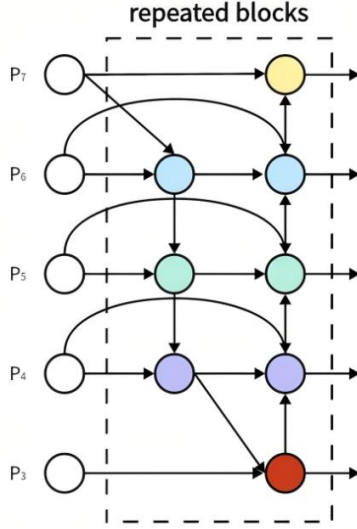


Figure 4. BiFPN structural diagram [21].

2. Bidirectional Connections: Unlike conventional FPN, BiFPN incorporates bidirectional links between adjacent levels of the feature pyramid. This allows information to flow from higher-level features to lower-level features (top-down) and from lower-level features to higher-level features (bottom-up).

3. Feature Integration: The bidirectional connections enable the integration of information from various levels of the feature pyramid in both directions, effectively capturing multi-scale features.

4. Weighted Feature Fusion: BiFPN utilizes a weighted feature fusion approach to merge features from different levels. The weights used for fusion are learned throughout the training process to optimize feature integration.

Because the input features come with different resolutions, their contributions to the output features are often unequal. To address this, we assign additional weights to each input, allowing the network to recognize the significance of each feature. The weights are represented as follows:

$$O = \sum_i \frac{w_i}{\epsilon + \sum_{jj}} \cdot I_i \quad (3)$$

This acceleration restriction method also solves the problems of slow speed and unstable training.

The final feature map combines bidirectional scale connections and fast normalization fusion, with P6 as a representative example:

$$P_6^{td} = \text{Conv}\left(\frac{w_1 \cdot P_6^{in} + w_2 \cdot \text{Resize}(P_7^{in})}{w_1 + w_2 + \epsilon}\right) \quad (4)$$

$$P_6^{out} = \text{Conv}\left(\frac{w_1' \cdot P_6^{in} + w_2' \cdot P_6^{td} + w_3' \cdot \text{Resize}(P_5^{out})}{w_1' + w_2' + w_3' + \epsilon}\right) \quad (5)$$

E. Counting method

Define the counter, which increases with the number of check boxes. Finally, call the cv2. putText() function to print the result on the picture or video in real time.

IV. EXPERIMENTS

A. Dataset

1) Data Collection: This study utilizes data from three distinct datasets: an online image dataset (referred to as the Internet dataset), a dataset from the JDD 2017 competition (known as the JD dataset), and a dataset collected from Guifeishan Farm of Guangxi Yangxiang Co., Ltd. (designated as the YX dataset).

2) Data Preprocessing: The Internet dataset was collected using Python web scraping techniques, and images with very low resolution or high similarity were manually removed. The JD dataset comprises footage of 30 pigs, each with approximately one minute of video. To form the dataset, several images were randomly extracted from the videos, with duplicates also being manually eliminated. The YX dataset consists of a long video recorded by a camera, which also captures random images of food while discarding those that are highly similar. This dataset includes clear images of pigs, as well as those showing issues like occlusion, thereby enhancing its diversity and complexity. This variety helps ensure that the trained model retains effective recognition capabilities in various challenging environments. Target samples were labeled in the images using the Labelling tool with rectangular bounding boxes. Below are examples of the representative image data included in this dataset.

B. Experimental environment and parameter settings

Variations in experimental environments and training parameters can influence the results. In this study, the experiments are relatively intensive. To maintain fairness, it is important to keep the experimental environment and training parameters consistent. The details of the experimental environment are presented in Table I. The training parameters included an initial learning rate of 0.01, a batch size of 12, 8 workers, and 300 epochs. No pre-trained models were utilized.

TABLE I. EXPERIMENTAL ENVIRONMENT

Category	Configuration
CPU	Inter(R) Core(TM)i7-9750H CPU @ 2.60GHz 2.59 GHz
GPU	NVIDIA GeForce GTX 1650
Framework	PyTorch 2.4.0
Language	Python3.8
Environment	PyCharm
Cuda version	11.8.0

C. Analysis of outcome indicators

The outcome indicator serves as a key criterion for evaluating the model's practicality. In this study, we focus on five specific metrics: accuracy (P), recall (R), Mean Average Precision (mAP), Params, and GFLOPs, to assess the model's performance. Accuracy measures the ratio of correctly identified positive samples within the model's recognition of evidence classes; a higher accuracy indicates superior model performance.

$$\text{Precision} = \frac{TP}{TP+FP} \quad (6)$$

The recall rate indicates the proportion of samples that the model accurately identifies as positive compared to the total number of actual positive samples. A higher recall signifies that the model correctly predicts more positive class samples, reflecting improved performance.

$$\text{Recall} = \frac{TP}{TP+FN} \quad (7)$$

Mean Average Precision (mAP) evaluates the overall performance across all categories by averaging the Average Precisions (APs) of each category. A higher mAP indicates better model performance.

Params denote the number of parameters within the model and are a key indicator of lightweight design. Given the requirement for accuracy, a lower number of parameters is preferable.

GFLOPs measures the computational complexity of the model and is crucial for lightweight considerations. A lower GFLOPs value signifies reduced computational complexity and a smaller model size.

D. Experimental comparison

To validate the effectiveness of the enhanced model, this study performed four comparative experiments: the original YOLOv5 model, YOLOv5 with the BiFPN module, YOLOv5 with the Ghost module, and YOLOv5 that incorporates all the improvements. The results of these experiments are presented in Table II.

TABLE II. EXPERIMENTAL RESULTS

Network	P (%)	R (%)	mAP (%)	Params (M)	GFLOPs
<i>YoLov5</i>	80.85	81.91	87.79	7.09	15.9
<i>YoLov5+BiFPN</i>	82.86	83.91	89.33	7.17	16.6
<i>YoLov5+Ghost</i>	79.72	82.44	86.95	4.80	11.6
<i>ours</i>	82.67	85.20	88.47	4.92	12.0

The table above clearly shows that, compared to the original YOLOv5 algorithm, introducing the BiFPN module alone enhances the mAP value, although it also increases both the parameters and computational complexity. On the other hand, while the Ghost module improves the algorithm's lightweight design, it results in a decline in accuracy.

However, when both the BiFPN and Ghost modules are used together, there is no significant increase in parameters or computational complexity, and the algorithm's accuracy improves. These experimental results indicate the effectiveness of this approach.

Additionally, we conducted an experiment by placing the Ghost module in the Backbone, but the results were not satisfactory. This may be because the Ghost module is a lightweight structure, and positioning it in the Backbone—an area critical for feature extraction—could lead to insufficient feature extraction due to the low-density convolution structure, causing significant dropout.

Next, we tested the model using recorded videos that were not part of the training set, extracting some images from the results as shown in Figure 5. Even when the pig was partially obstructed, it was still accurately identified.

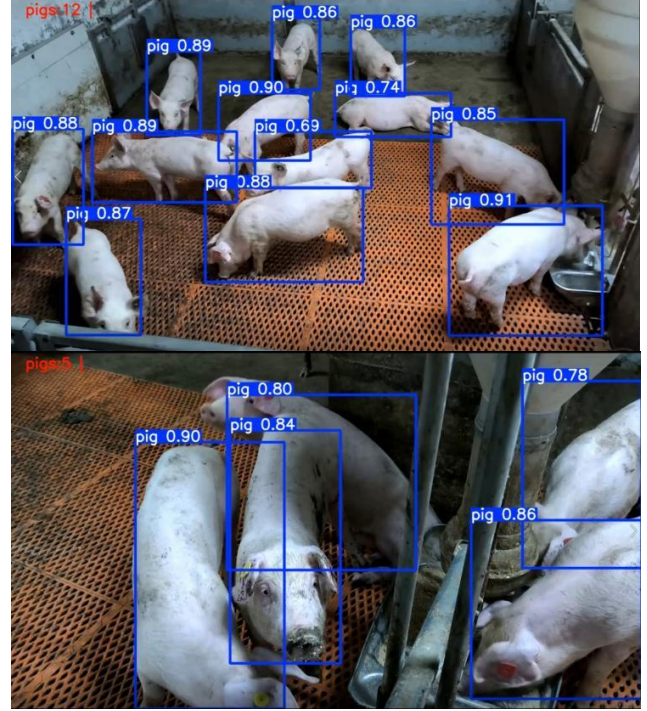


Figure 5. Test results.

V. CONCLUSION

This article presents an enhanced method for pig object detection based on YOLOv5, which increases recognition accuracy while maintaining a compact model size. The proposed improvements to YOLOv5 include the following: incorporating the Ghost module, which utilizes smaller filters to generate additional feature maps from the original convolutional layer, resulting in improved recognition performance; adding the BiFPN module, which leverages bidirectional connections to better capture feature representations at various scales, enhancing the network's ability to process objects of different sizes and complexities, thereby improving detection robustness and accuracy; and implementing a counting function to display the number of pigs detected for easier counting. Experimental results

indicate that this approach has yielded favorable outcomes and fulfills the deployment and real-time detection requirements of YOLOv5. However, there are still some limitations. For instance, while enhancing each module individually proved effective, integrating multiple modules did not achieve optimal performance. Future research will focus on further optimizing the model and exploring additional application areas.

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